Machine learning for predicting sediment particle size distributions

UE24CS352A Machine Learning · Mini-project 62 · Team B-16 · Bandugula Sai Rithesh Reddy (PES1UG24AM064), Gagana C (PES1UG24AM099)

1. Problem statement

Mud particles suspended in estuaries stick together into flocs and break apart again within seconds to minutes. Their size sets how fast they settle, which is the most uncertain input to sediment-transport, erosion and contaminant models. No general physical model predicts floc size well. Following Egan (Stanford CS229, 2019), we ask whether a regression model can predict two summaries of the particle size distribution (PSD) from routinely measured water properties: the **median diameter d50** and the **spread σ** (the volume-weighted standard deviation of diameter, i.e. the square root of the paper's variance σ^2). Success is measured, as in the paper, by the coefficient of determination R^2 on held-out data.

2. Dataset

The paper's own data are not public, so we used the closest public equivalent: the U.S. Geological Survey release Profiles of salinity, temperature, depth, turbidity, and distributions of particle size in suspension … south San Francisco Bay, summer 2020 (doi:10.5066/P99Q4CHM). It covers the same bay with the same instrument family (a Sequoia LISST-200X laser particle sizer) plus an RBR CTD with a turbidity sensor.

- **Coverage:** vertical casts at a deep channel site (11–14 m) and a shallows site (2–5 m) on four days (21, 22, 24, 28 July 2020). The package was held about a minute at each depth.
- **Raw size:** 94 LISST files (45,639 one-second samples after removing one file that duplicates another) and 99 CTD files sampled six times a second. The LISST gives volume concentration in 36 size classes (1.2–459 µm), temperature and beam attenuation; the CTD gives salinity, turbidity and depth.
- **After cleaning:** 3,998 ten-second bursts from 92 casts (50 channel, 42 shallows).
- **Compared with the paper:** we have its strongest predictor, salinity, and temperature, but not chlorophyll, refractive index or wave and tidal velocity.

3. Approach

Targets. We recompute d50 and σ from each burst-averaged PSD; our d50 agrees with the value stored in the files ($r = 0.98$). **Features (9).** Sensor depth, height above the bed, total water depth (the tidal stage), its rate of change between casts (flood or ebb), temperature, salinity, turbidity, beam attenuation (a second proxy for sediment concentration) and a site flag. Total volume concentration was deliberately left out because it is computed from the PSD itself.

Models. We reproduce the paper's two models, Random Forest and support vector regression, and add a ridge baseline, gradient boosting and a 10-unit neural network (the architecture of Sahin et al., 2017). As in the paper, feature importance is averaged over repeated 10-tree forests, the forest size (50 trees) is chosen where the out-of-bag score levels off, and SVR's C and ε come from a validation sweep ($C = 0.5$, $\varepsilon = 0.25$ on standardised targets).

Evaluation: the main extension. The paper closes by asking whether its models would work at another time or place. Consecutive samples in a cast are nearly identical, so a random split puts near-copies of every test point in the training set. We therefore report four protocols of increasing difficulty: (1) the paper's random 70/30 split, (2) five-fold cross-validation with whole casts held out, (3) leave-one-day-out, and (4) train on one site, test on the other.

4. Implementation

- **src/build_dataset.py** reads the NetCDF casts and decodes the EPIC time format. The two instruments keep separate clocks, so it finds the 3–7 s offset each day by lining up their depth traces, then joins them second by second. It drops in-air samples and samples outside 30–98 % optical transmission (the LISST's valid range), and averages into 10-second bursts taken at a steady depth. Depth comes from the CTD, whose pressure sensor proved more reliable.
- **src/psd.py** computes d50 (log-interpolated on the cumulative volume curve) and σ.
- **src/train_evaluate.py** runs feature importance, tuning, all five models under all four protocols and a feature-set ablation; **src/plots.py** draws the figures.
- **demo.py** holds out one cast, retrains and prints predicted against measured values, or predicts for conditions typed on the command line. Python 3, scikit-learn, xarray; fixed seed.

5. Results

R² on test data for each model and protocol (mean over splits; higher is better, below 0 is worse than predicting the mean).

Model	Random 70/30 (paper)		Unseen casts		Unseen day		Unseen site	
	d50	σ	d50	σ	d50	σ	d50	σ
Ridge (linear baseline)	0.43	0.53	0.36	0.45	0.19	0.31	-5.39	< −10
Random Forest	0.86	0.82	0.56	0.58	0.35	0.46	-1.67	-0.95
SVR (RBF)	0.69	0.72	0.60	0.62	0.45	0.53	-1.90	-3.83
Gradient Boosting	0.85	0.81	0.59	0.60	0.42	0.50	-2.32	-1.17
Neural net (10 units)	0.69	0.72	0.53	0.62	0.34	0.52	< −10	-7.09

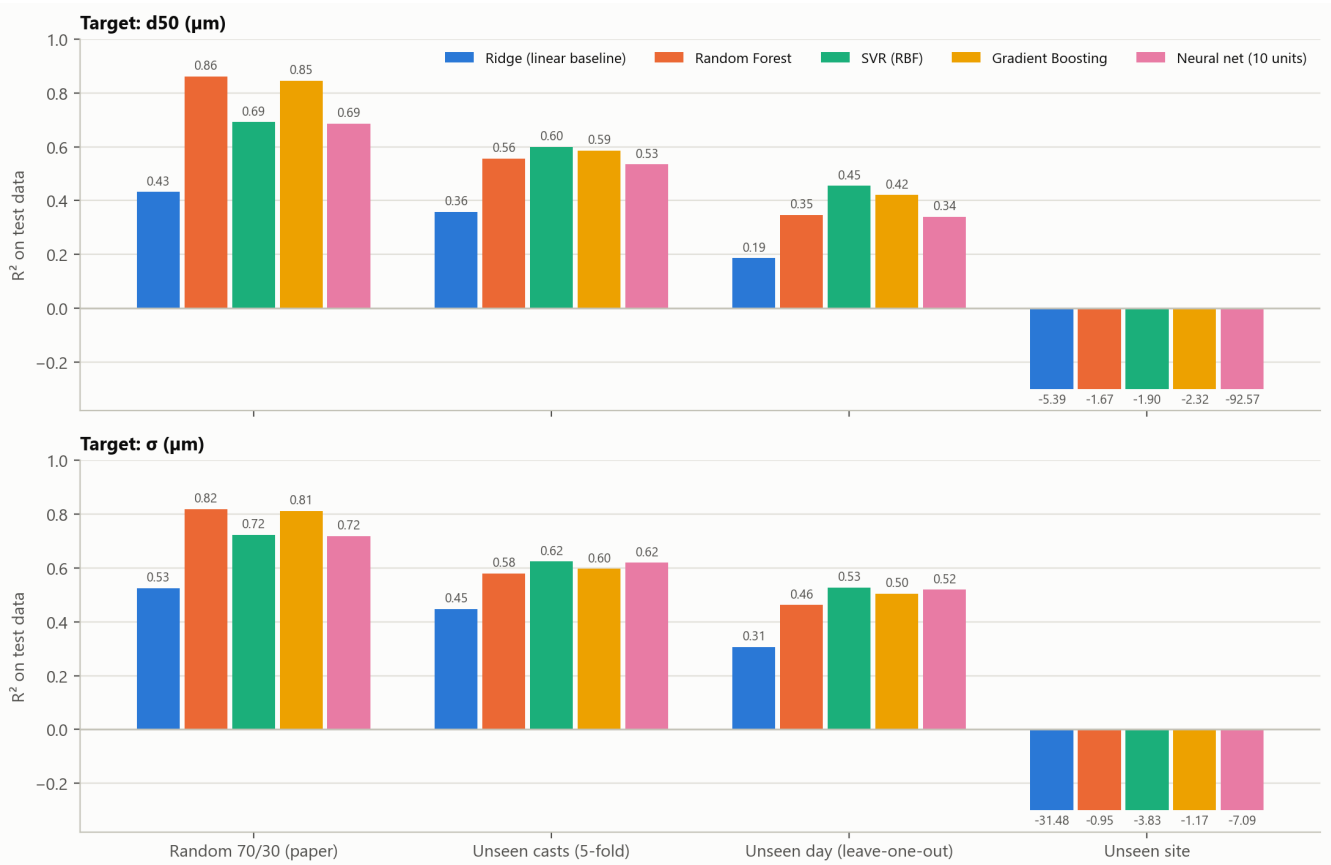


Figure 1. R² of every model under the four test protocols. Bars are cut off at −0.3; labels give the true value.

- **The paper's result reproduces.** With a random split, Random Forest reaches R² = 0.86 for d50 and 0.82 for σ; the paper reports 0.81 and 0.75 on different data and features.
- **Part of that score is memorised neighbours.** On casts the model never saw, Random Forest falls to 0.56 and 0.58. SVR, the smoother model, is now best (0.60 and 0.62); on an unseen day it scores 0.46 and 0.53.
- **Nothing transfers between sites.** Every model has negative R² when trained on one site and tested on the other, because water depth and particle populations barely overlap.
- **What drives the predictions.** Sensor depth, water depth (tide stage) and beam attenuation matter most. Salinity, which dominated in the paper (importance 0.72 for d50), carries only 8 % and 7 % here: over our four days it stayed between 30.1 and 32.3 psu. With Random Forest on unseen casts, position, tide and temperature give R² = 0.42 / 0.46 (d50 / σ); adding beam attenuation gives 0.54 / 0.58; adding salinity and turbidity gives 0.56 / 0.58.

6. Conclusions

From depth, tide, temperature, salinity and two optical measurements, machine learning explains about 56–62 % of the variation in floc size and spread on new casts. That is useful, but short of the 0.82–0.86 a random split suggests. The headline lesson is methodological: for autocorrelated field data the split decides the score, and the paper's worry about overfitting was justified. Limits of our study: four days in one summer, two sites, a very narrow salinity range, and beam attenuation measured by the same instrument as the targets. The next step is to add current and wave measurements from the release's hydrodynamic time series and to train across more sites and seasons.